Racial Achievement Gaps in Wisconsin Public Schools

A Decomposition Analysis with Focus on the Madison Metropolitan School District

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Abstract

Wisconsin has the largest Black-White achievement gap in fourth-grade reading in the nation. Using Wisconsin Department of Public Instruction Forward Exam data (2015–2023), we document statewide racial proficiency gaps, decompose them into within- and between-district components, and conduct a deep-dive analysis of the Madison Metropolitan School District (MMSD). We find that the statewide Black-White gap is approximately 38–40 percentage points in ELA and 42–44 percentage points in mathematics, and has been stable across the entire sample period. About 47% of the Black-White gap is between school districts and 53% is within districts. Within MMSD, a surprising finding emerges: MMSD Black students (8–10% proficient in ELA) perform at lower rates than Black students in Milwaukee and other peer districts, despite MMSD's higher overall district reputation. Furthermore, 82% of the MMSD Black-White gap exists within individual schools — not between schools. This final result has direct implications for MMSD's ongoing attendance boundary review: even optimal boundary redesign can address only the 18% between-school component of the gap, while the majority of the racial disparity would remain unaffected by student reassignment alone.

Introduction

Wisconsin's racial achievement gap is among the most severe in the nation. On the 2024 National Assessment of Educational Progress (NAEP), Wisconsin posted the largest Black-White gap in fourth-grade reading nationwide, exceeding states historically associated with school segregation such as Louisiana, Mississippi, and South Carolina (Reardon et al. 2019). On the Wisconsin Forward Exam — the state's primary accountability assessment for grades 3–8 — the Black-White proficiency gap stands at approximately 43 percentage points, with Black students proficient at a rate of 18% compared to 61% for White students (Wisconsin Department of Public Instruction, n.d.-a).

This paper is motivated by a concrete policy question: the Madison Metropolitan School District (MMSD) is conducting a two-year attendance boundary review (2025–2027) with the stated goal of promoting "socio-economic, linguistic, and racial diversity within and across schools" (Madison Metropolitan School District 2025). To assess the potential impact of boundary changes on racial achievement gaps, we need to know two things: how large is MMSD's racial gap, and how much of that gap is between schools (and thus potentially addressable by boundary changes) versus within schools (and thus unaffected by student reassignment).

We address these questions with three complementary analyses. First, we document statewide racial proficiency gaps by race, grade, and subject over the primary Forward Exam window (2015–16 to 2022–23, excluding COVID-disrupted years). Second, we decompose the statewide Black-White and Hispanic-White gaps into within- and between-district components using a standard linear decomposition (Reardon et al. 2024). Third, we conduct a detailed analysis of MMSD, including: (a) comparing MMSD minority students to same-race peers in peer districts with substantial minority enrollment (Milwaukee, Racine, Kenosha, Green Bay, and others); (b) comparing MMSD minority students to White students in non-MMSD Wisconsin districts, to account for the outlier-SES status of MMSD White students; and (c) decomposing

the MMSD gap into within- and between-school components.

Our main findings are as follows. The statewide BW gap is large (~38–44 pp depending on subject), persistent (no trend toward closure 2015–2023), and approximately equally split between within- and between-district components. Within MMSD, however, the within-school component dominates: 82% of the MMSD BW gap exists within individual schools, and only 18% is between schools. Perhaps most strikingly, MMSD Black students perform at lower proficiency rates than Black students in Milwaukee and several other peer districts — undermining the common assumption that minority students in MMSD benefit from their district's overall high reputation.

The paper proceeds as follows. Section reviews the literature. Section describes the data and analytical approach. Section documents statewide gaps. Section presents the within-between decomposition. Section presents the MMSD-specific analysis. Section discusses policy implications.

Background

The National Literature on Racial Achievement Gaps

Racial achievement gaps are a persistent feature of American public education. Using data from the Stanford Education Data Archive (SEDA) — which links state accountability tests from approximately 200 million students to a common NAEP scale — Reardon et al. (2019) document that Black-White achievement gaps vary enormously across school districts, ranging from near zero to greater than 1.5 standard deviations. Economic, demographic, segregation, and schooling characteristics explain 43–72% of this geographic variation. The strongest correlates are racial differences in parental income and education, and patterns of racial school segregation.

More recent work using SEDA data through 2019 finds that Black-White gaps have been

growing modestly, not converging (Matheny et al. 2023). There is no evidence that districts that improve overall achievement simultaneously narrow racial gaps — suggesting that general academic improvement is not sufficient to reduce racial disparities.

The mechanism linking school segregation to achievement gaps has been clarified by Reardon et al. (2024), who find that racial segregation affects gaps primarily through "racial economic segregation" — concentrating minority students in high-poverty schools. After controlling for racial differences in school poverty exposure, racial segregation has little independent predictive power. This finding implies that the relevant policy lever is not racial mixing per se but reducing minority students' exposure to concentrated poverty.

Wisconsin's Exceptional Gap

Wisconsin's position at the top of the national BW gap distribution has been documented since at least the early 2000s using WKCE (Wisconsin Knowledge and Concepts Exam) data and subsequently on the Forward Exam. The Wisconsin Institute for Law & Liberty (2026) policy report attributes approximately 42% of the school-level relationship between Black student share and proficiency to poverty rates, and an additional 3.6% to disability identification rates. The remaining gap — after controlling for poverty and disability — is attributed to unmeasured factors including family structure and early literacy practices, though these claims rest on cross-sectional ecological data and should be interpreted with caution.

The MMSD Context and the Boundary Review

MMSD serves approximately 26,000 students across 51 elementary schools and is one of the most racially diverse districts in Wisconsin outside of Milwaukee. The district's White student population is an outlier by state standards: Madison is a college town anchored by the University of Wisconsin-Madison, and MMSD White families have unusually high parental

education and income. This demographic fact has important methodological implications: the MMSD within-district BW gap is partly a reflection of unusually advantaged White students rather than unusually disadvantaged Black students.

The ongoing "Building for the Future" boundary review (Madison Metropolitan School District 2025) is being conducted with third-party vendor MGT and will evaluate enrollment trends, programming, and diversity outcomes through 2027. The boundary review raises the question of whether and how much student reassignment across MMSD schools could reduce the racial achievement gap. The answer depends critically on how much of the gap is between schools (potentially addressable by boundary changes) versus within schools (unaffected by student reassignment).

The research literature on boundary redrawing provides important context. Gillani et al. (2023) simulated alternative attendance boundaries for 98 US elementary districts and found a median 14% relative decrease in White/non-White segregation achievable under optimal algorithmic redesign, while requiring approximately 20% of students to switch schools. At the district level, Sorensen et al. (2024) find that while redrawing attendance zones within a district reduces segregation by less than 5% in typical cases, redrawing school district boundaries could reduce more than 40% of segregation. These findings frame our decomposition analysis: the boundary review has the potential to affect only the between-school component of the gap.

Data and Methods

Data Sources

Forward Exam data. Our primary data source is Wisconsin DPI Forward Exam assessment data for school years 2015–16 through 2024–25, downloaded from the DPI assessment portal (Wisconsin Department of Public Instruction, n.d.-a). The Forward Exam covers grades 3–8

in ELA and Mathematics. The data are reported in long format at the school × grade × subject × race × proficiency level. We aggregate to school-level and district-level proficiency rates (Proficient + Advanced) by race group.

WKCE historical data. For historical context predating the Forward Exam, we use Wisconsin Knowledge and Concepts Exam (WKCE) Scale Score Summary (SSS) files for 2003–04 through 2013–14 (Wisconsin Department of Public Instruction, n.d.-b). The WKCE era provides mean scale scores and percentile distributions by race at the district level. Results from the two assessment eras are reported separately and are not directly compared.

Analysis window. We use two primary analysis windows:

Window	Years	Rationale
Primary	2015-16, 2016-17, 2017-18, 2018-19, 2021-22, 2022-23	Same standards, no COVID disruption
New standards	2023-24, 2024-25	Cut scores revised; not comparable to prior years

The 2019-20 school year had no testing (federal COVID waiver). The 2020-21 school year had low and non-representative participation and is excluded from the primary analysis.

Suppression

DPI suppresses results for race groups with fewer than 10 tested students. Suppressed cells are coded as NaN and excluded from all analyses. We report suppression rates throughout the paper. At the district level (our primary unit for the decomposition), Black student suppression is approximately 18%, making district-level analysis feasible for nearly all large and mid-size districts.

Decomposition Method

We decompose the statewide Black-White gap into within- and between-district components using the following linear decomposition. Let p_w^d, p_b^d denote White and Black proficiency rates in district d, and n_w^d, n_b^d denote the corresponding counts of tested students. Then:

$$T = \bar{p}_w - \bar{p}_b = \sum_d \frac{n_w^d}{N_w} p_w^d - \sum_d \frac{n_b^d}{N_b} p_b^d$$

where $N_w = \sum_d n_w^d$ and $N_b = \sum_d n_b^d$. The between-district component B captures the gap attributable to racial groups attending districts with systematically different average performance:

$$B = \sum_d \bar{p}_d \left(\frac{n_w^d}{N_w} - \frac{n_b^d}{N_b} \right), \quad \text{where } \bar{p}_d = \frac{n_w^d p_w^d + n_b^d p_b^d}{n_w^d + n_b^d}$$

The within-district component $W = T - B$ captures the enrollment-weighted average within-district gap. By construction, $T = B + W$ exactly. We apply this decomposition at three levels of geographic aggregation: district, county, and school.

Peer District Comparison

For the MMSD analysis, we identify a set of peer districts with sufficient Black and Hispanic enrollment for reliable year-over-year comparisons. Using the criterion of at least 30 tested students in a given race group in at least 5 primary years, we find 82 eligible districts statewide. We designate three tiers for the peer comparison:

Tier	Districts
Tier 1 — major urban	Milwaukee, Racine Unified, Kenosha, Green Bay Area Public, Beloit

Tier	Districts
Tier 2 — mid-size	Sun Prairie Area, Appleton Area, Waukesha, Janesville, West Allis-West Milwaukee
Madison region	Verona Area, Middleton-Cross Plains Area

Statewide Racial Proficiency Gaps

Gap Magnitudes

Figure 1 shows enrollment-weighted statewide proficiency gaps between White and Black students, and between White and Hispanic students, for each primary year. Thin lines represent individual grades (3–8); the bold line is the grade-enrollment-weighted average.

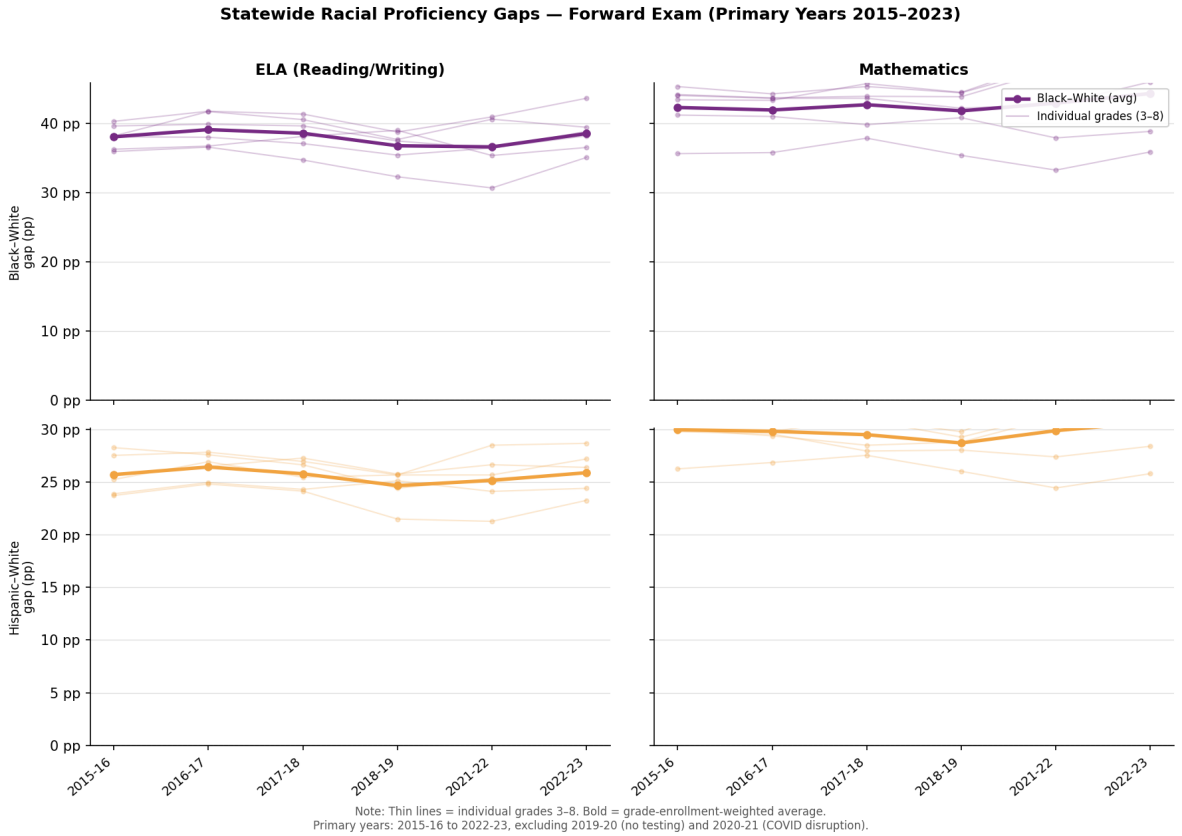

Figure 1: Statewide Racial Proficiency Gaps over Time — Forward Exam Primary Years 2015–2023

Table 3: Statewide Proficiency Gaps by Year and Subject (Grade-Weighted, Primary Years)

School Year	BW ELA	BW Math	HW ELA	HW Math
2015-16	38.1	42.6	25.7	30.1
2016-17	39.2	42.2	26.5	29.9
2017-18	38.7	43	25.8	29.6
2018-19	36.9	42.2	24.8	28.9
2021-22	37	44.1	25.4	30.5
2022-23	38.8	45.3	26	31.1

Note: Thin lines = individual grades 3–8; bold line = grade-enrollment-weighted average. Primary years exclude 2019-20 (no testing) and 2020-21 (COVID disruption).

The statewide Black-White gap averages approximately 38–40 percentage points in ELA and 42–44 percentage points in Mathematics across primary years (see Table 3). The Hispanic-White gap is approximately 10 percentage points smaller: 25–26 pp in ELA and 29–31 pp in Mathematics. Math gaps exceed ELA gaps for both comparisons.

Gap Stability

A striking feature of Figure 1 is the near-complete stability of the BW and HW gaps across the entire 2015–2023 primary window. Neither gap shows a meaningful trend toward closure. This is consistent with national evidence from Matheny et al. (2023), who find that BW gaps grew slightly in the 2009–2019 period nationally, and with the general conclusion that improvements in overall district achievement do not translate into gap reduction.

Gaps by Grade

Figure 2 shows gaps pooled across primary years by grade.

The BW gap is broadly similar across grades 3–8, ranging from approximately 36 to 42 pp in ELA and 40 to 46 pp in Math, with no consistent pattern of widening or narrowing across the elementary-middle school transition.

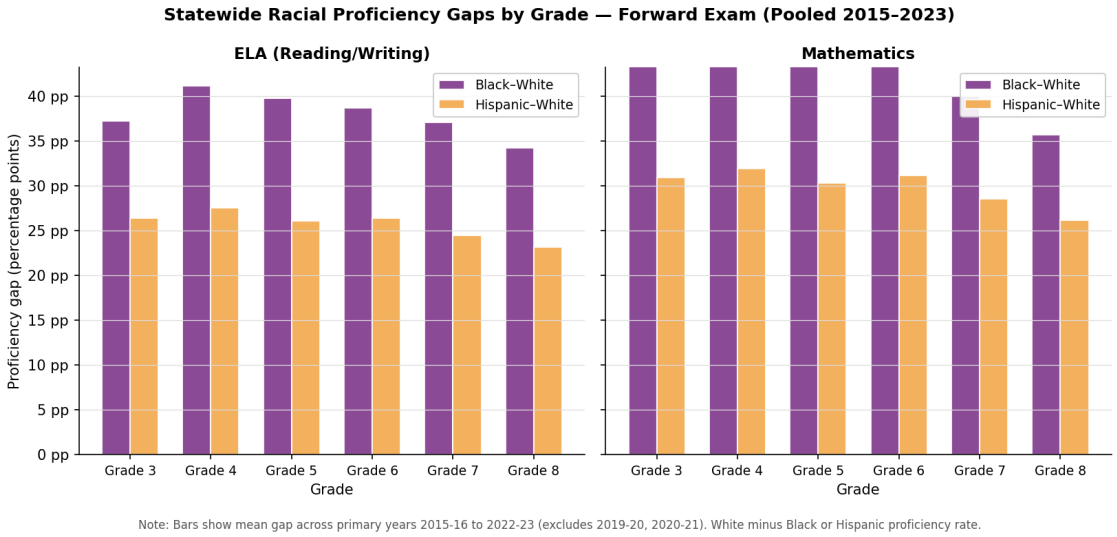

Figure 2: Proficiency Gaps by Grade — Pooled Across Primary Years

Within-Between Decomposition

District-Level Results

Figure 3 shows the within-between decomposition of the BW and HW gaps at the district, county, and school levels.

At the district level, approximately 47% of the BW gap is between districts (explained by White and Black students attending different districts with different average performance) and 53% is within districts (gaps that exist even within the same district). For the HW gap, the split is more skewed toward within districts: approximately 35% between, 65% within. These results are stable across all primary years.

The district-level decomposition implies that even if students were perfectly distributed across districts — eliminating all between-district variation — the statewide BW gap would still be approximately 20 percentage points (the within component). District-level boundary changes, while potentially valuable for integration, cannot eliminate the gap by themselves.

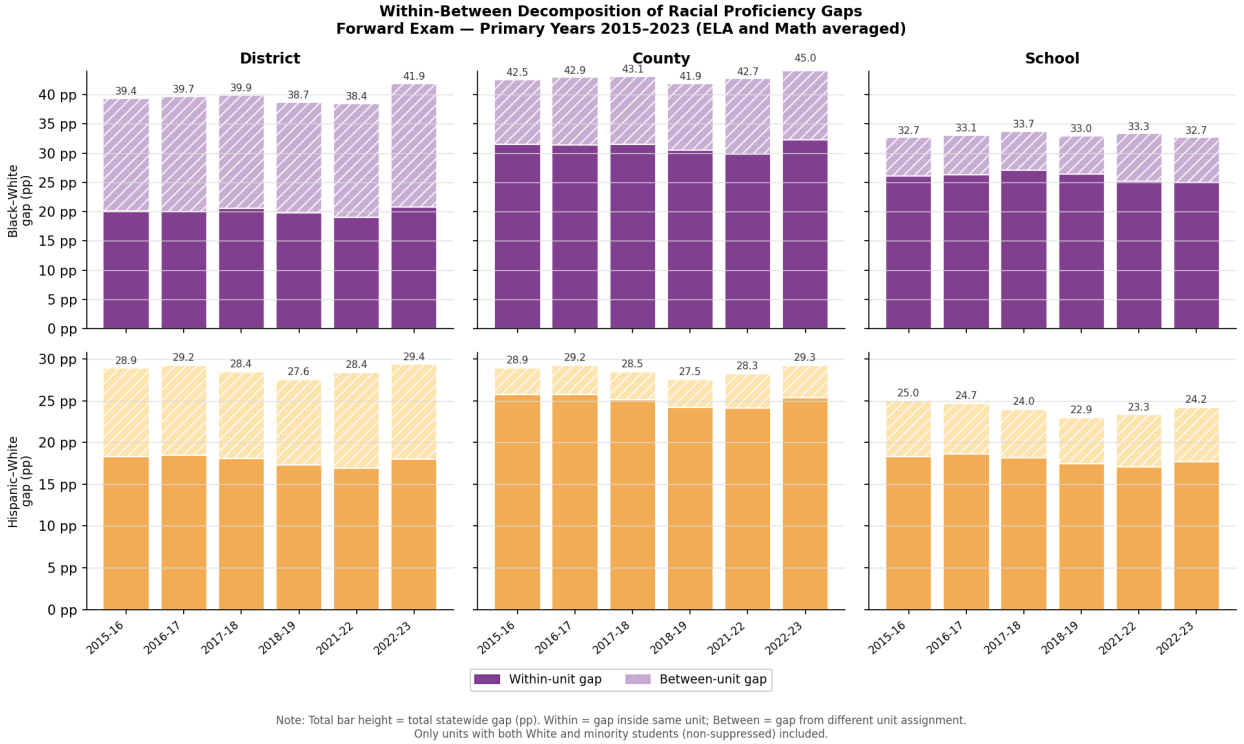

Figure 3: Within-Between Decomposition of Racial Proficiency Gaps

Relationship Between School Districts

Figure 4 shows the district-level scatterplot of Black vs. White ELA proficiency, averaged across primary years.

Each point above the parity line (dashed) represents a district where White students outperform Black students. MMSD (labeled in red) sits far above the parity line, consistent with an unusually large within-district gap. The scatter also shows that Black student proficiency varies substantially across districts — ranging from near zero to above 30% in ELA — while White student proficiency is more compressed.

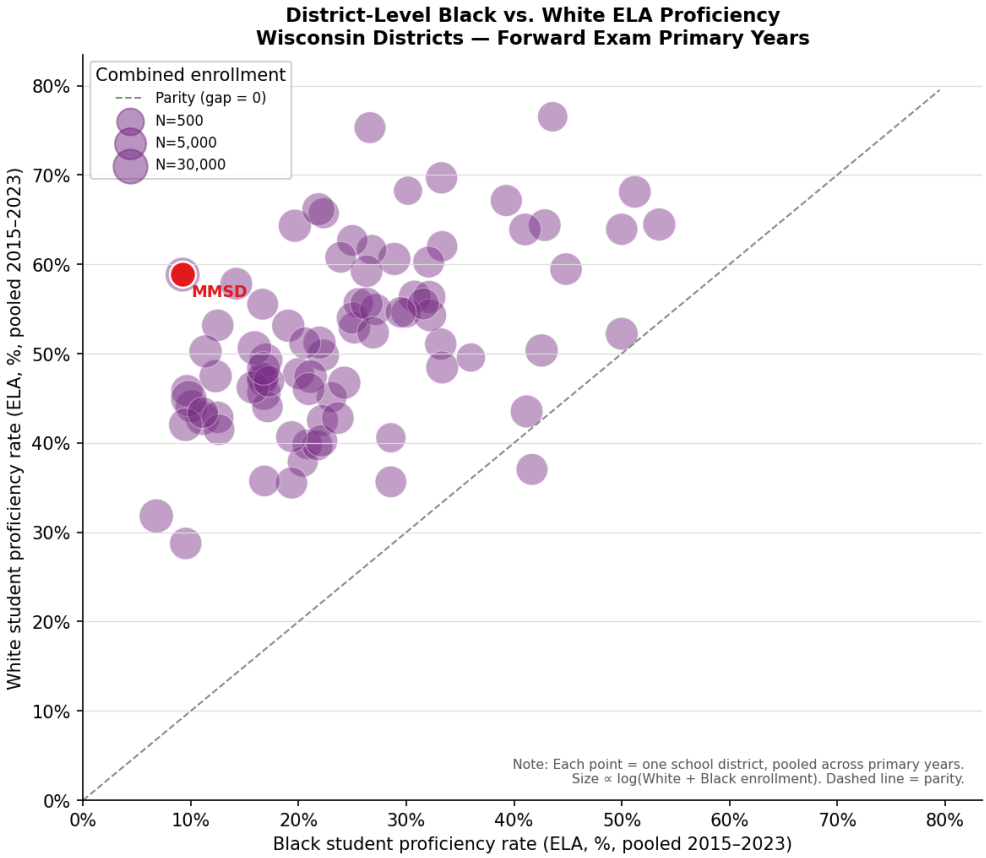

Figure 4: District-Level Black vs. White ELA Proficiency (Pooled 2015–2023)

MMSD Analysis

MMSD Minority Students vs. Peer Districts

A natural question is whether MMSD minority students perform well relative to their same-race peers in other districts. The commonly assumed story is that MMSD, as a high-performing district, provides better outcomes for all students including minorities. Figure 5 challenges this assumption.

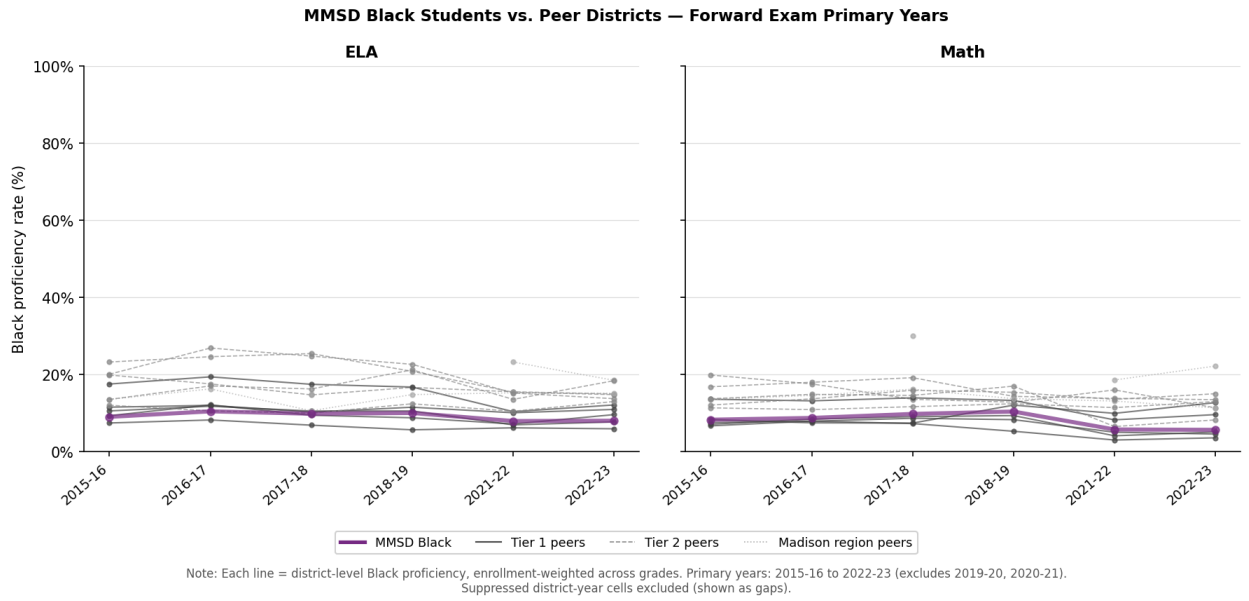

Figure 5: MMSD Black Students vs. Peer District Black Students

MMSD Black students achieve ELA proficiency rates of approximately 8–10% across primary years. This is substantially below Milwaukee Black students (approximately 16–22%) and below several Tier 2 peer districts as well. The same pattern holds in Mathematics. Figure 6 shows a similar pattern for Hispanic students: MMSD Hispanic proficiency (~15–17%) is broadly comparable to or below peer districts.

These findings imply that the large within-MMSD racial gap is not primarily the result of minority students performing unusually well in MMSD — rather, it reflects the unusually high performance of MMSD White students combined with minority students performing no

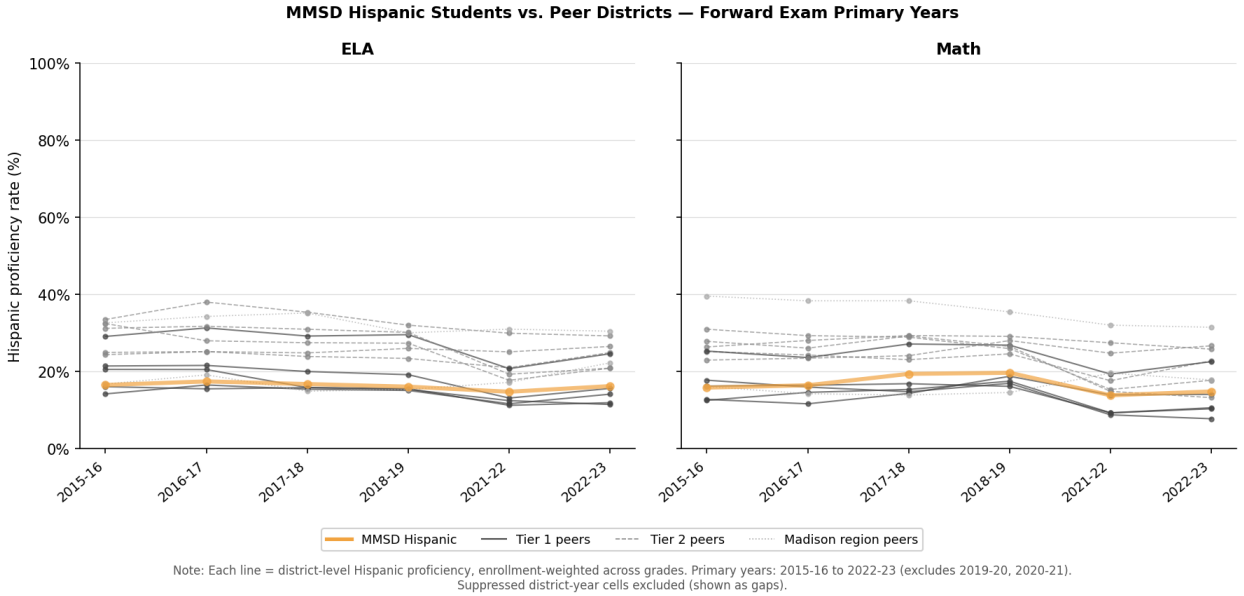

Figure 6: MMSD Hispanic Students vs. Peer District Hispanic Students

better (and in some dimensions, worse) than in peer districts.

MMSD Minority vs. Non-MMSD Wisconsin White

To further underscore the outlier-SES status of MMSD White students, Figure 7 compares MMSD Black and Hispanic proficiency directly to the proficiency of White students in all Wisconsin districts except MMSD.

Non-MMSD Wisconsin White students achieve ELA proficiency rates of approximately 45–53%, while MMSD Black students achieve only 8–10% and MMSD Hispanic students achieve 15–17%. The gap between MMSD Black students and non-MMSD Wisconsin White students is approximately 40–43 percentage points — comparable to the statewide Black-White gap reported in Section . This comparison illustrates that MMSD minority students are performing substantially below even the statewide White average, not merely below the exceptional MMSD White population.

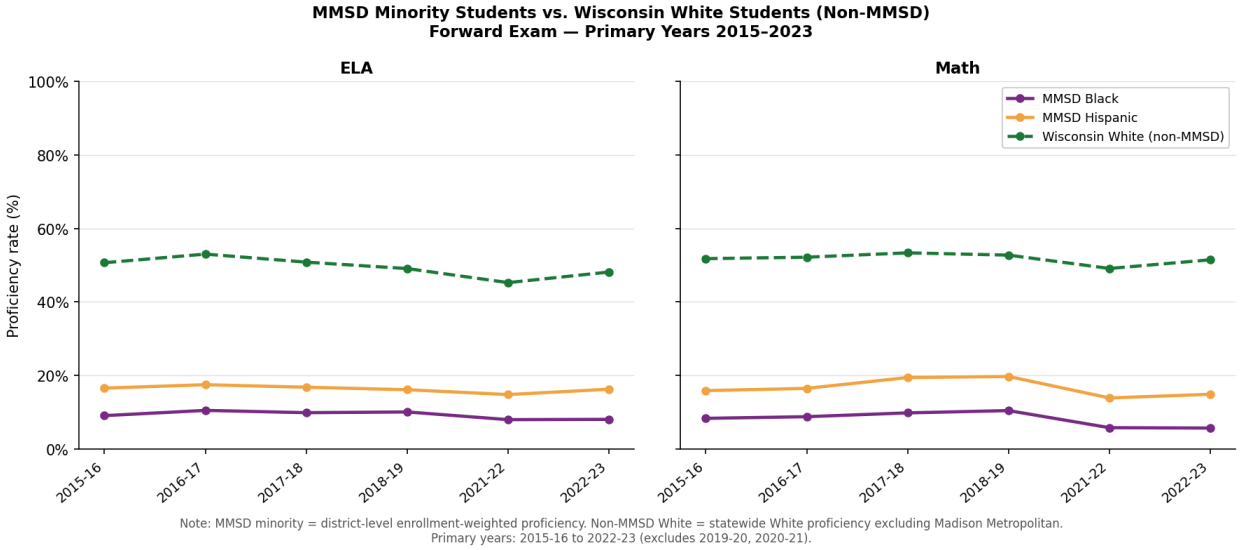

Figure 7: MMSD Minority Students vs. Non-MMSD Wisconsin White Students

Within-MMSD School Distribution

Figure 8 shows the distribution of school-level proficiency within MMSD by race and year, pooled across grade and subject.

Note: Each box represents the distribution of school-level proficiency rates for that race group across MMSD's elementary schools, for non-suppressed schools only.

Several features of this figure are noteworthy. First, the level of Black proficiency in MMSD schools is uniformly low — the median MMSD school has Black ELA proficiency of approximately 5–12%, and there is no MMSD school where Black students approach the White proficiency rate. Second, the within-school decomposition reveals that approximately 82% of the MMSD BW gap is within individual schools, and only 18% is between schools.

This decomposition is the central policy-relevant finding of this paper. The MMSD attendance boundary review could, in principle, redistribute students across schools and thereby reduce the between-school component of the gap. Under the most optimistic assumptions — complete equalization of racial composition across schools, comparable to the best-case simulations in Gillani et al. (2023) — the boundary review would reduce the MMSD BW gap by

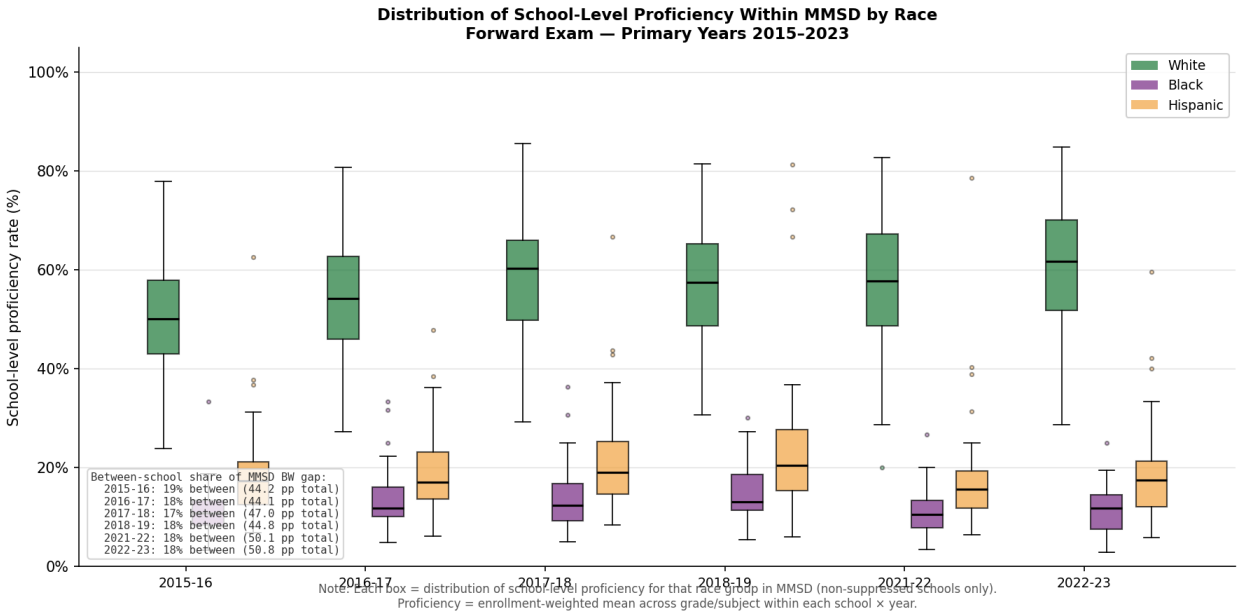

Figure 8: Distribution of School-Level Proficiency Within MMSD by Race

approximately 14% × 18% 2.5 percentage points out of a roughly 47 pp within-MMSD gap. The gap would remain approximately 44 pp.

Conclusion

Wisconsin's racial achievement gap is large, national in rank, and persistent over time. Our analysis documents a statewide Black-White proficiency gap of approximately 38–44 pp on the Forward Exam (depending on subject), with roughly half attributable to between-district sorting and half within districts.

For MMSD specifically, three findings stand out. First, MMSD Black and Hispanic students perform no better than — and in some cases worse than — same-race students in Milwaukee and other peer districts, despite MMSD's higher overall district ranking. This reflects the outlier-SES composition of MMSD White students, not unusually good outcomes for minority students. Second, the gap between MMSD minority students and non-MMSD Wisconsin White students (approximately 40 pp in ELA) is nearly as large as the statewide average BW

gap, underlining that MMSD minority students are struggling relative to a representative White comparison group, not just relative to the extraordinary MMSD White population. Third, and most directly relevant to the boundary review, 82% of the MMSD BW gap exists within individual schools. Redistricting can address only the 18% between-school component.

These findings do not imply that boundary changes are without value — school integration has other benefits beyond direct test score effects, including exposure to diverse peers and reduced concentration of poverty in individual schools (Reardon et al. 2024). But they do suggest that boundary changes alone are unlikely to produce large reductions in the racial achievement gap. Policies targeting the within-school component of the gap — instructional quality, early intervention, teacher distribution, and the school poverty concentration that underlies the gap — would need to be central to any comprehensive equity strategy.

Data Availability

All analysis code, data download scripts, and figure-generation scripts are available at [GitHub repository link]. Raw data are from Wisconsin DPI and are publicly available via WISEdash.

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Appendix

A. Suppression Rates by Race and Level

[Table to be added: suppression rates at district level by race × year]

B. Decomposition Results — Full Table

[Table to be added: full T, B, W, B_pct, W_pct by level × year × subject × gap pair]

C. COVID Sensitivity Check

[Table to be added: key results including 2020-21, showing robustness of main findings]